

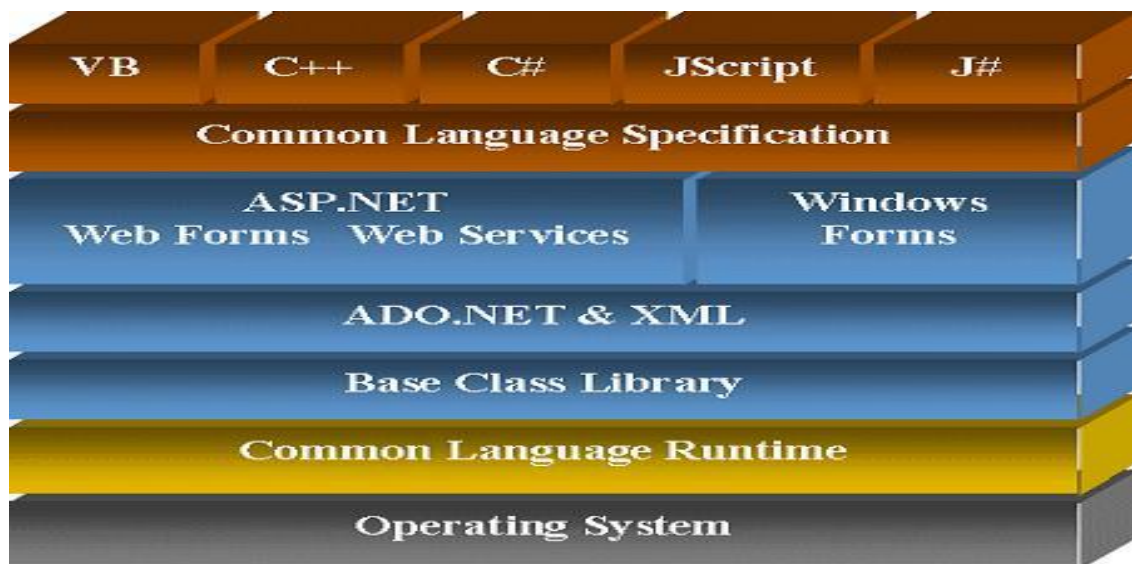
UNIT 4

DOT NET FRAMEWORK

[4Hrs]

4.1. Introduction, Common Language Runtime (CLR):**4.1.1. Introduction:**

- The first version of the .Net framework was released in the year 2002.
- The .NET Framework is a revolutionary platform created by Microsoft for developing applications.
- The .NET Framework enables the creation of **Windows applications, Web applications, Web services**, and pretty much anything else we can think of.
- The .NET Framework has been designed so that it can be used from any language, including C# as well as C++, Visual Basic, JScript, and even older languages such as COBOL.

**Fig: .Net Framework Architecture****4.1.2. Common Language Runtime (CLR):**

- The Common Language Runtime (CLR) is the execution engine that handles running applications.
- The runtime layer is referred as CLR.
- The primary role of the CLR is to **locate, load, and manage .NET types** on our behalf.
- The CLR also takes care of a number of low-level details such as memory management, application hosting, handling threads, and performing various security checks.
- CLR allows for the easy use of different supported languages to achieve a common goal.

4.2.Common Type System (CTS):

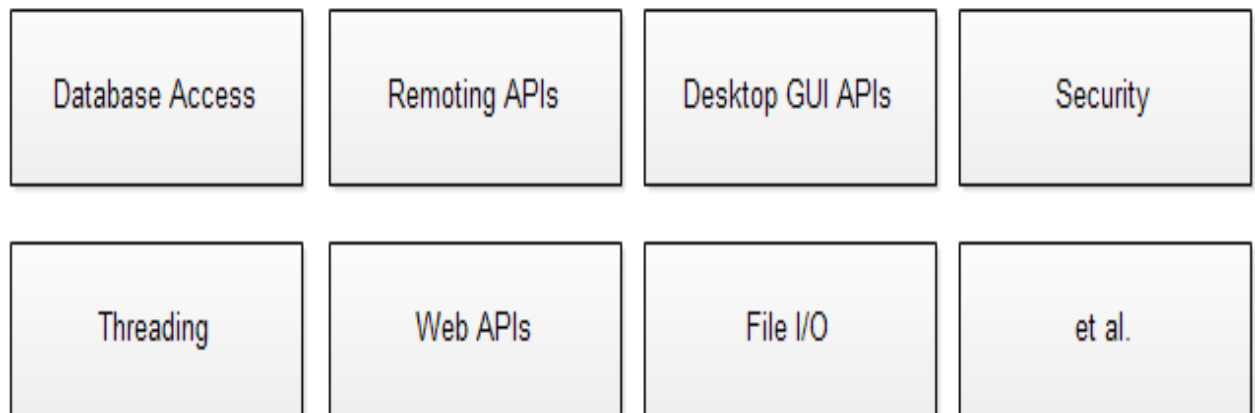
- The Common Type System (CTS) is a standard for defining and using data types in the .NET framework.
- CTS defines a collection of data types, which are used and managed by the run time to facilitate cross-language integration.
- Establishes a framework that helps enable cross-language integration, type safety, and high-performance code execution.
- Provides an object-oriented model that supports the complete implementation of many programming languages.
- Defines rules that languages must follow, which helps ensure that objects written in different languages can interact with each other.

4.3.Common Language Specification (CLS):

- Common Language Specification (CLS) is a set of basic language features that .Net Languages needed to develop Applications and Services, which are compatible with the .Net Framework.
- CLS defines the base rules necessary for any language targeting common language infrastructure to interoperate with other CLS-compliant languages.
- CLS represents the guidelines to the compiler of a language, which targets the .NET Framework
- CLS includes features common to many object-oriented programming languages. It forms a subset of the functionality of common type system (CTS) and has more rules than defined in CTS.

4.4.The base class library, The .net class library intermediate language:

- Base class library also known as Class library (CL).
- BCL is a subset of Framework class library (FCL).
- Class library is the collection of reusable types that are closely integrated with CLR.
- Base Class library provides classes and types that are helpful in performing day to day operation e.g. dealing with string and primitive types, database connection, IO operations.
- BCL define types that can be used to build any type of software application.

**Fig:Base Class Library (BCL)**

4.5. Just in Time (JIT) and Common Intermediate Language compilation, garbage collection:

4.5.1. CIL and JIT:

- CIL (Common Intermediate Language) is a bytecode and the language of the .NET platform into which source code written in high level languages are compiled rather than creating direct native code.
- Its operations are based on a stack and it is executed by a virtual machine.
- It is also called IL or MSIL (Microsoft Intermediate Language) which was its original name before the standardization of the CLI (Common Language Infrastructure) which it belongs to and with which it should not be confused.
- The source code from high-level language (C#, Basic or other language) is compiled into CIL and stored into an assembly.
- This code is compiled into bytecode to be interpreted by a JIT, or binary to be executed directly by the processor.

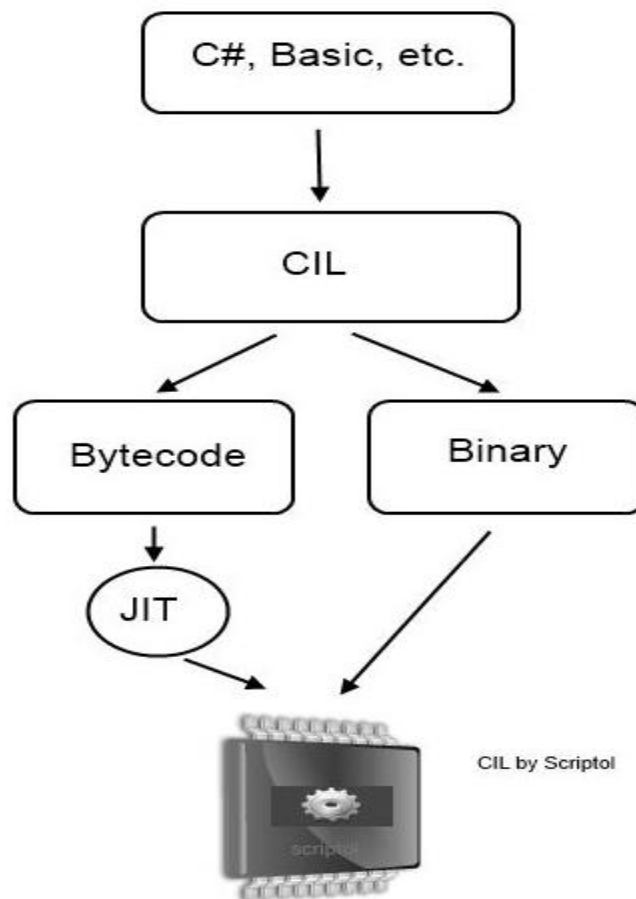
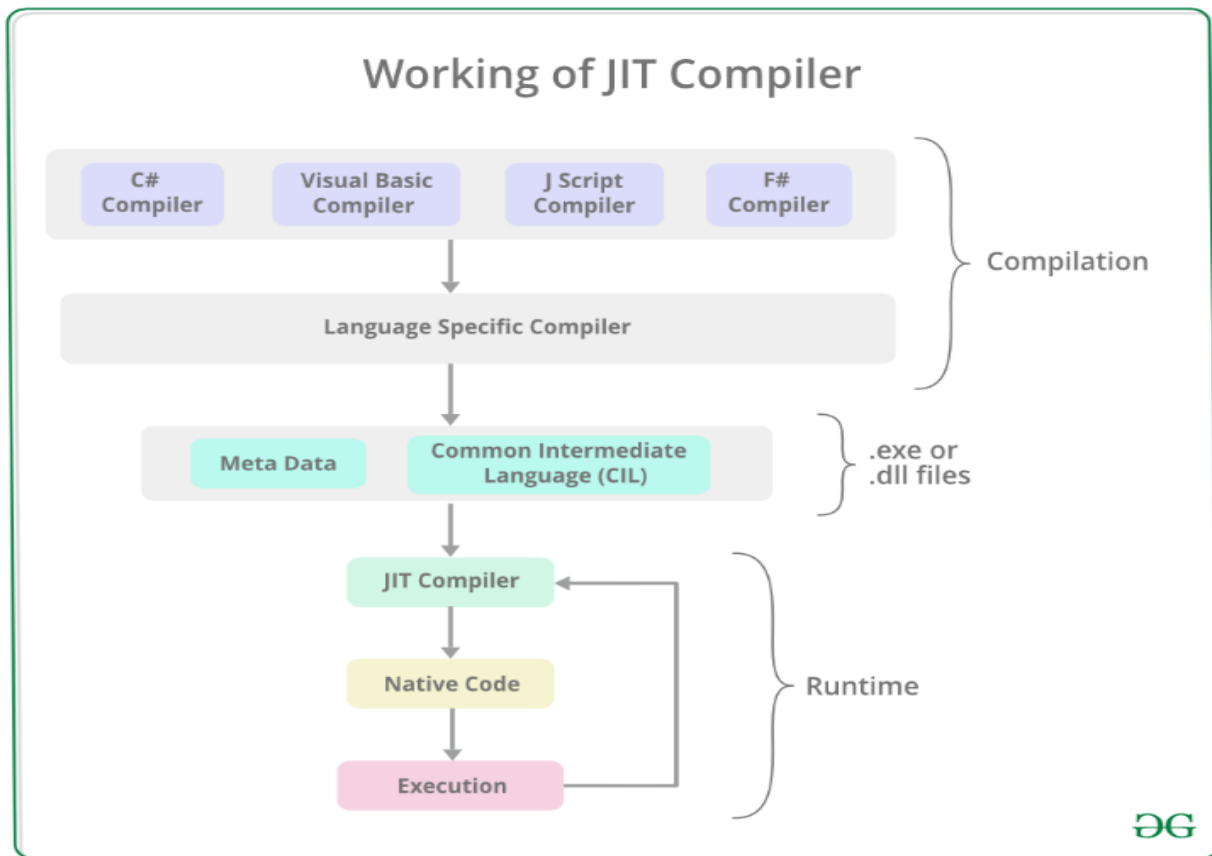


Fig: Working of CIL

❖ JIT:

- Just-In-Time compiler(JIT) is a part of Common Language Runtime (CLR) in .NET which is responsible for managing the execution of .NET programs regardless of any .NET programming language.
- This intermediate language is converted into the machine code by the Just-In-Time (JIT) compiler.

**Fig: Working of JIT****Advantages of JIT Compiler:**

- The JIT compiler requires less memory usage as only the methods that are required at run-time are compiled into machine code by the JIT Compiler.
- Page faults are reduced by using the JIT compiler as the methods required together are most probably in the same memory page.
- Code optimization based on statistical analysis can be performed by the JIT compiler while the code is running.

Disadvantages of JIT compiler:

- The JIT compiler requires more startup time while the application is executed initially.
- The cache memory is heavily used by the JIT compiler to store the source code methods that are required at run-time.

❖ **Managed and Unmanaged Code:**

MANAGED CODE	UNMANAGED CODE
It is executed by managed runtime environment or managed by the CLR.	It is executed directly by the operating system.
It provides security to the application written in .NET Framework.	It does not provide any security to the application.
Memory buffer overflow does not occur.	Memory buffer overflow may occur.
It provides runtime services like Garbage Collection, exception handling, etc.	It does not provide runtime services like Garbage Collection, exception handling, etc.
The source code is compiled in the intermediate language known as <i>IL</i> or <i>MSIL</i> or <i>CIL</i> .	The source code directly compiles into native language.
It does not provide low-level access to the programmer.	It provides low-level access to the programmer.

4.5.2. Garbage Collection:

- Automatic memory management is made possible by Garbage Collection in .NET Framework.
- It is the method of freeing the memory space used by an application when the application is no longer in use.
- Garbage collection will always work on Managed Heap and internally it has an Engine which is known as the Optimization Engine.
- Garbage Collection occurs if at least one of multiple conditions is satisfied. These conditions are given as follows:
 - ✓ If the system has low physical memory, then garbage collection is necessary.
 - ✓ If the memory allocated to various objects in the heap memory exceeds a pre-set threshold, then garbage collection occurs.
 - ✓ If the GC. Collect method is called, then garbage collection occurs. However, this method is only called under unusual situations as normally garbage collector runs automatically.

Phases in Garbage Collection

- Marking Phase: List of live Objects Created
- Relocating Phase: References for list of live objects updated
- Compacting Phase: Dead objects released, live objects compacted and moved

4.6. Application installation and Assemblies:**Summary of steps required to create a .NET application**

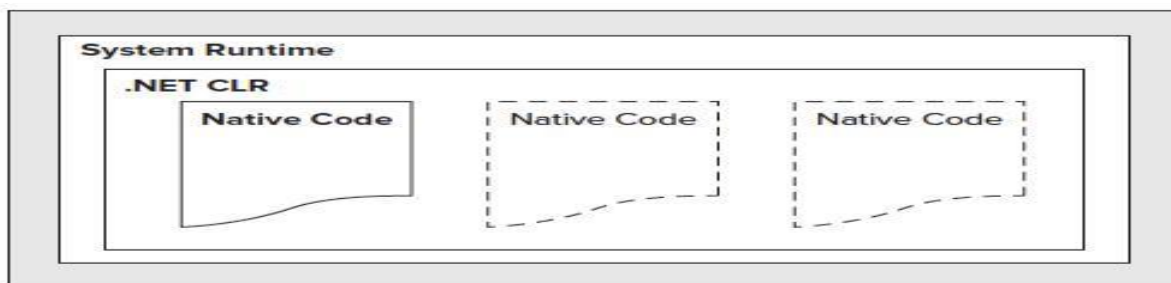
1. Application code is written using a .NET-compatible language such as C#.
2. That code is compiled into CIL, which is stored in an assembly.



3. When this code is executed (either in its own right if it is an executable or when it is used from other code), it must first be compiled into native code using a JIT compiler.



4. The native code is executed in the context of the managed CLR, along with any other running applications or processes.

**❖ Assemblies:**

- Assemblies are the collection of both executable application files and libraries for the use of other application.
- When we compile an application, the CIL code created is stored in an *assembly*.
- All the .NET assemblies contain the definition of types, versioning information for the type meta-data, and manifest.
- An assembly is a collection of types and resources that are built to work together and form a logical unit of functionality.
- Assemblies take the form of executable (.exe) or dynamic link library (.dll) files, and are the building blocks of .NET applications.

4.7.Windows Communication Foundation (WCF):

- Windows Communication Foundation is just that the foundation for communication between Windows computers.
- It just so happens that thanks to open standards like SOAP and ReST, WCF can communicate with other software systems.
- It is a framework for building service-oriented applications.
- Using WCF, you can send data as asynchronous messages from one service endpoint to another.
- A service endpoint can be part of a continuously available service hosted by IIS, or it can be a service hosted in an application.

Features of WCF:

WCF includes the following set of features.

1. Service Orientation
2. Interoperability
3. Multiple Message Patterns
4. Security
5. Multiple Transports and Encodings
6. Reliable and Queued Messages
7. Durable Messages
8. Transactions

4.8.Windows Presentation Foundation (WPF):

- WPF, or Windows Presentation Foundation, is a graphical system for rendering user interfaces.
- It provides great flexibility in how we can lay out and interact with our applications.
- The advantages of WPF for our application are its rich data binding and visualization support and its design flexibility and styling.
- WPF enables us to create an application that is more usable to our audience.
- It gives us the power to design an application that would previously take extremely long development cycles and a calculus genius to implement.
- Windows Presentation Foundation (WPF) is a UI framework that creates desktop client applications.
- The WPF development platform supports a broad set of application development features, including an application model, resources, controls, graphics, layout, data binding, documents, and security.

Features of WPF are as following:

- Extensible Application Markup Language (XAML)
- Controls
- Text, media, document
- Layout
- 2D and 3D graphics
- Animation
- Styles
- Templates etc.

Important Questions from this Unit:

- Q1) Explain about CIL and JIT in detail with block diagram?
- Q2) How the dot net applications are developed? Explain each step briefly.
- Q3) Explain about WCF and WPF and CTS?